Full results for medium-sized datasets

Overview datasets

Dataset	n	d	Type	Source
Abalone	4,176	10	Tabular	UCI ML Repository
Travel	5,454	24	Tabular	UCI ML Repository
Facebook	7,050	13	Tabular	UCI ML Repository
Frogs	7,194	22	Tabular	UCI ML Repository
Electric	10,000	12	Tabular	UCI ML Repository
Npi	10,440	40	Tabular	Online personality tests
Pulsar	17,898	8	Tabular	UCI ML Repository
Creditcard	30,000	24	Tabular	UCI ML Repository
Adult	32,560	110	Tabular	UCI ML Repository
Plants	34,780	70	Tabular	UCI ML Repository
Bank	45,210	53	Tabular	UCI ML Repository
Cifar10	50,000	3,072	Image	UCI ML Repository
Mnist	60,000	784	Image	Kaggle

Metric: Cityblock (L1, Manhattan)

Objective function values

					Deviation from best objective (%)										
Dataset	n	d	Best objective	UB gap (%)	GRB	HEX-B	HEX-L	BLV	ABM	ABM-i1	ABM-i2	ABM-d5000	AC	GRD	RND
Abalone	4,176	10	32,155.09	-0.02	0.00	0.00	0.19	0.00	0.01	0.00	0.00	0.01	0.04	0.13	31.28
Travel	5,454	24	86,060.14	-1.12	0.00	0.19	3.32	0.00	0.14	0.05	0.03	0.14	0.52	2.98	24.28
Facebook	7,050	13	29,806.77	0.00	0.00	0.00	0.01	0.00	0.00	0.00	0.00	0.00	0.00	11.69	19.84
Frogs	7,194	22	116,081.63	-1.28	0.00	0.31	2.99	0.00	0.12	0.05	0.03	0.12	0.47	4.19	27.18
Electric	10,000	12	103,133.34	-0.19	0.00	—	4.45	0.00	0.24	0.08	0.04	0.24	0.87	0.94	32.94
Npi	10,440	40	134,708.00	0.00	0.00	—	4.98	0.00	0.00	0.00	0.00	0.25	2.04	7.18	35.23
Pulsar	17,898	8	82,970.56	-0.03	0.00	—	1.33	—	0.00	0.00	0.00	0.01	0.01	2.00	23.39
Creditcard	30,000	24	407,488.30	-0.07	—	—	5.15	—	0.01	0.00	0.00	0.07	—	2.54	26.01
Adult	32,560	110	760,952.87	0.00	—	—	2.09	—	0.00	0.00	0.00	0.00	—	1.26	17.17
Plants	34,780	70	302,103.00	0.00	—	—	0.07	—	0.00	0.00	0.00	0.00	—	3.64	14.51
Bank	45,210	53	950,153.07	0.00	—	—	4.38	—	0.00	0.00	0.00	0.00	—	—	22.30
Cifar10	50,000	3,072	29,223,842.46	-1.22	—	—	22.77	—	0.23	0.06	0.00	1.18	—	—	25.27
Mnist	60,000	784	5,165,664.04	-2.63	—	—	18.42	—	0.20	0.05	0.00	1.17	—	—	24.12
Average					—	—	5.40	—	0.07	0.02	0.01	0.25	—	—	24.89

Runtimes (seconds)

			Runtime (s)										
Dataset	n	d	GRB	HEX-B	HEX-L	BLV	ABM	ABM-i1	ABM-i2	ABM-d5000	AC	GRD	RND
Abalone	4,176	10	23.89	4,581.71	7,200.33	976.22	1.71	3.33	5.06	1.75	23.44	8.60	<0.01
Travel	5,454	24	53.34	7,321.13	7,200.91	806.55	2.65	5.10	7.62	2.75	52.84	14.93	<0.01
Facebook	7,050	13	81.96	11,434.70	7,201.11	2,334.97	7.91	17.57	27.53	8.03	111.53	23.39	<0.01
Frogs	7,194	22	101.15	7,929.18	7,200.89	3,832.08	5.95	11.41	16.85	5.99	118.01	25.22	<0.01
Electric	10,000	12	257.12	14,400.01	7,201.03	1,710.92	6.99	13.21	18.85	7.04	305.74	57.13	<0.01
Npi	10,440	40	315.86	14,400.01	7,202.93	1,798.88	6.26	11.31	15.93	5.81	350.01	44.38	<0.01
Pulsar	17,898	8	1,120.04	14,400.02	7,201.06	—	139.62	275.73	413.40	26.48	2,055.67	196.02	<0.01
Creditcard	30,000	24	—	1,756.95	7,205.99	—	608.19	1,274.77	1,933.95	25.61	—	774.54	<0.01
Adult	32,560	110	—	2,298.28	7,224.97	—	1,313.15	2,789.78	4,270.34	43.21	—	1,063.46	<0.01
Plants	34,780	70	—	2,580.46	7,215.95	—	88.36	121.44	153.24	2.80	—	2,080.88	<0.01
Bank	45,210	53	—	2,347.56	7,216.02	—	1,914.01	3,456.40	5,083.33	25.08	—	—	<0.01
Cifar10	50,000	3,072	—	7,962.81	8,225.49	—	1,474.37	2,851.99	4,153.33	38.49	—	—	<0.01
Mnist	60,000	784	—	2,160.49	7,493.16	—	1,456.33	2,831.96	4,110.46	17.66	—	—	<0.01
Average			—	7,197.95	7,306.91	—	540.42	1,051.08	1,554.61	16.21	—	—	<0.01

Metric: Euclidean (L2)

Objective function values

					Deviation from best objective (%)										
Dataset	n	d	Best objective	UB gap (%)	GRB	HEX-B	HEX-L	BLV	ABM	ABM-i1	ABM-i2	ABM-d5000	AC	GRD	RND
Abalone	4,176	10	11,545.25	-0.85	0.00	0.00	0.19	0.00	0.04	0.02	0.01	0.04	0.12	1.09	27.68
Travel	5,454	24	22,720.98	-1.64	0.00	0.25	0.75	0.00	0.14	0.06	0.03	0.14	0.50	3.61	18.79
Facebook	7,050	13	15,446.27	-1.19	0.00	0.00	4.72	0.00	0.00	0.00	0.00	0.00	0.01	22.40	20.23
Frogs	7,194	22	29,392.31	-1.73	0.00	0.42	1.23	0.00	0.14	0.07	0.04	0.14	0.56	5.69	24.75
Electric	10,000	12	33,298.77	-0.11	0.00	—	2.63	0.00	0.51	0.26	0.16	0.51	1.41	1.36	27.83
Npi	10,440	40	26,448.15	-0.70	0.00	—	2.77	0.00	0.08	0.03	0.02	0.22	1.13	5.75	20.66
Pulsar	17,898	8	37,448.93	-0.95	0.00	—	5.10	—	0.03	0.01	0.01	0.08	0.12	7.75	23.27
Creditcard	30,000	24	117,382.42	-1.55	—	—	4.55	—	0.07	0.02	0.00	0.27	—	7.16	21.28
Adult	32,560	110	234,124.50	-2.03	—	—	2.94	—	0.01	0.00	0.00	0.06	—	7.02	8.96
Plants	34,780	70	66,986.71	-2.01	—	—	2.89	—	0.00	0.00	0.00	0.00	—	11.67	12.06
Bank	45,210	53	254,976.81	-2.16	—	—	3.13	—	0.02	0.00	0.00	0.12	—	—	11.93
Cifar10	50,000	3,072	607,744.99	-1.06	—	—	19.61	—	0.21	0.06	0.00	1.04	—	—	21.91
Mnist	60,000	784	361,609.21	-0.83	—	—	11.69	—	0.12	0.03	0.00	0.69	—	—	15.66
Average					—	—	4.79	—	0.10	0.04	0.02	0.25	—	—	19.62

Runtimes (seconds)

			Runtime (s)										
Dataset	n	d	GRB	HEX-B	HEX-L	BLV	ABM	ABM-i1	ABM-i2	ABM-d5000	AC	GRD	RND
Abalone	4,176	10	31.77	4,650.84	7,200.24	1,106.10	1.33	2.57	3.76	1.37	23.41	8.55	<0.01
Travel	5,454	24	55.07	7,322.99	7,200.83	778.97	2.65	5.01	7.27	2.72	52.62	14.89	<0.01
Facebook	7,050	13	99.50	11,213.13	7,201.12	5,180.55	8.51	18.28	27.51	8.67	110.50	22.89	<0.01
Frogs	7,194	22	102.70	7,911.40	7,200.86	4,054.78	6.01	11.51	17.02	6.14	117.97	27.80	<0.01
Electric	10,000	12	150.45	14,400.01	7,201.04	511.33	6.62	11.81	16.49	6.69	307.62	56.07	<0.01
Npi	10,440	40	216.84	14,400.01	7,202.95	6,070.69	8.18	16.47	25.04	8.25	349.95	43.21	<0.01
Pulsar	17,898	8	1,006.84	14,400.02	7,201.05	—	135.64	263.36	393.06	26.52	2,062.87	196.27	<0.01
Creditcard	30,000	24	—	1,752.49	7,206.01	—	463.70	904.10	1,332.34	21.32	—	785.17	<0.01
Adult	32,560	110	—	2,462.84	7,224.93	—	1,031.25	1,962.37	2,881.27	36.97	—	3,046.44	<0.01
Plants	34,780	70	—	2,627.11	7,215.95	—	1,045.95	1,877.47	2,847.16	31.05	—	1,666.16	<0.01
Bank	45,210	53	—	2,368.11	7,215.94	—	2,009.44	3,692.06	5,330.54	28.06	—	—	<0.01
Cifar10	50,000	3,072	—	6,375.53	8,220.00	—	1,456.08	2,749.87	3,978.58	33.23	—	—	<0.01
Mnist	60,000	784	—	1,953.08	7,492.71	—	1,413.45	2,707.36	3,917.77	18.07	—	—	<0.01
Average			—	7,064.43	7,306.43	—	583.75	1,094.02	1,598.29	17.62	—	—	<0.01

Metric: Squared Euclidean (L2 squared)

Objective function values

					Deviation from best objective (%)											
Dataset	n	d	Best objective	UB gap (%)	GRB	HEX-B	HEX-L	BLV	ABM	ABM-i1	ABM-i2	ABM-d5000	AC	FAC-500	GRD	RND
Abalone	4,176	10	74,817.50	-0.87	0.00	0.00	0.07	0.00	0.06	0.03	0.02	0.06	0.19	0.48	0.77	44.20
Travel	5,454	24	193,609.69	-2.08	0.00	0.44	0.79	0.00	0.27	0.11	0.06	0.27	0.93	2.52	4.37	32.06
Facebook	7,050	13	105,777.75	-5.47	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.75	12.91
Frogs	7,194	22	255,910.88	-2.62	0.00	0.84	1.11	0.00	0.32	0.15	0.09	0.32	0.95	2.39	6.23	38.33
Electric	10,000	12	225,348.23	-0.22	0.00	—	5.01	0.00	1.06	0.52	0.33	1.06	2.72	8.25	2.01	46.86
Npi	10,440	40	134,708.00	-21.24	0.00	—	5.09	0.00	0.00	0.00	0.00	0.26	2.04	5.61	7.18	35.23
Pulsar	17,898	8	218,298.67	-5.34	0.00	—	1.05	—	0.07	0.03	0.02	0.17	0.20	1.01	4.42	34.16
Creditcard	30,000	24	1,057,515.25	-2.98	—	—	5.27	—	0.09	0.02	0.00	0.38	—	2.36	6.31	31.97
Adult	32,560	110	3,967,987.47	-0.30	—	—	1.89	—	0.01	0.00	0.00	0.05	—	—	1.41	9.73
Plants	34,780	70	302,103.00	-1.07	—	—	0.07	—	0.00	0.00	0.00	0.00	—	—	3.64	14.51
Bank	45,210	53	2,942,646.16	-2.92	—	—	3.92	—	0.02	0.00	0.00	0.13	—	—	—	18.53
Cifar10	50,000	3,072	15,766,598.45	-0.80	—	—	35.79	—	0.43	0.12	0.00	2.14	—	—	—	39.52
Mnist	60,000	784	4,383,206.85	-1.78	—	—	21.36	—	0.24	0.06	0.00	1.38	—	—	—	27.87
Average					—	—	6.26	—	0.20	0.08	0.04	0.48	—	—	—	29.68

Runtimes (seconds)

			Runtime (s)											
Dataset	n	d	GRB	HEX-B	HEX-L	BLV	ABM	ABM-i1	ABM-i2	ABM-d5000	AC	FAC-500	GRD	RND
Abalone	4,176	10	34.47	4,619.92	7,200.69	879.83	1.26	2.43	3.59	1.30	69.17	0.33	8.57	<0.01
Travel	5,454	24	52.47	7,322.19	7,200.82	603.96	2.50	4.74	6.90	2.53	283.23	0.46	14.93	<0.01
Facebook	7,050	13	98.17	11,213.84	7,201.05	5,770.09	8.46	17.73	27.45	8.55	417.22	0.48	25.46	<0.01
Frogs	7,194	22	107.53	7,876.87	7,200.91	3,499.80	5.44	10.35	15.12	5.49	621.44	0.56	27.65	<0.01
Electric	10,000	12	171.24	—	7,201.06	525.30	5.68	10.32	14.48	5.75	1,064.19	0.74	57.42	<0.01
Npi	10,440	40	313.31	—	7,202.94	1,802.36	6.24	11.22	15.86	5.67	3,039.43	1.00	43.04	<0.01
Pulsar	17,898	8	1,094.56	—	7,201.04	—	99.47	189.26	277.99	19.57	4,446.20	1.22	195.99	<0.01
Creditcard	30,000	24	—	—	7,205.98	—	445.08	894.32	1,339.09	20.16	—	2.20	783.79	<0.01
Adult	32,560	110	—	—	7,224.94	—	1,121.07	2,221.40	3,321.13	38.10	—	—	2,968.28	<0.01
Plants	34,780	70	—	—	7,215.93	—	90.63	122.71	154.62	2.86	—	—	1,958.71	<0.01
Bank	45,210	53	—	—	7,216.05	—	2,155.42	3,922.25	5,687.82	28.46	—	—	—	<0.01
Cifar10	50,000	3,072	—	—	8,237.64	—	1,196.64	2,250.01	3,241.17	30.89	—	—	—	<0.01
Mnist	60,000	784	—	—	7,497.86	—	1,327.40	2,598.93	3,815.22	17.56	—	—	—	<0.01
Average			—	—	7,308.22	—	497.33	942.74	1,378.50	14.38	—	—	—	<0.01